Apache Cassandra Overview

Sure! Let's dive into the topic of Apache Cassandra in simple terms.

Apache Cassandra is like a super-efficient library that can store a huge number of books (data) and make them available to readers (users) at lightning speed. Imagine a library where every book is stored in a way that you can find it quickly, even if there are thousands of other books around. This library is designed to handle many visitors at once, ensuring that everyone can get the information they need without waiting in line. It’s built to be very reliable, so even if some parts of the library are closed for maintenance, the rest of the library is still open and accessible.

One of the coolest things about Cassandra is that it can grow easily. If more people start visiting the library, you can simply add more shelves (servers) without having to reorganize everything. This makes it perfect for popular services like Netflix or Uber, where lots of people are constantly accessing data. So, if you think of data storage like a library, Apache Cassandra is the ultimate library that never closes, always has room for more books, and helps everyone find what they need quickly!

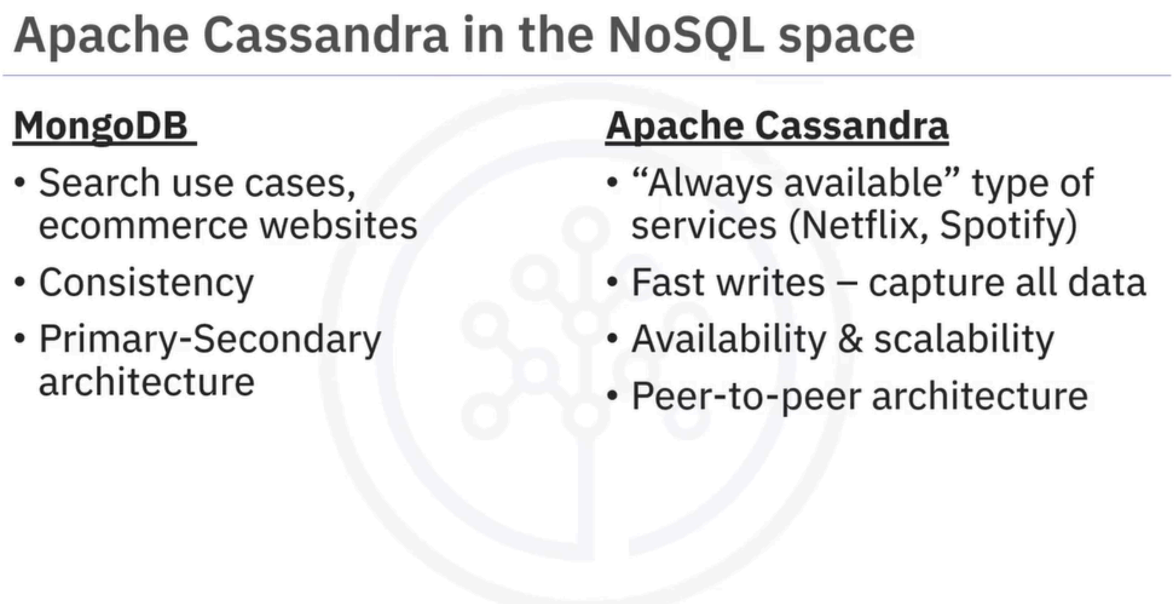
What is Apache Cassandra?

"Apache Cassandra is an ***open source, distributed, decentralized, elastically scalable, highly available, fault-tolerant, tunable, and consistent database*** that bases its distribution design on Amazon's Dynamo and its data model on Google's Bigtable.

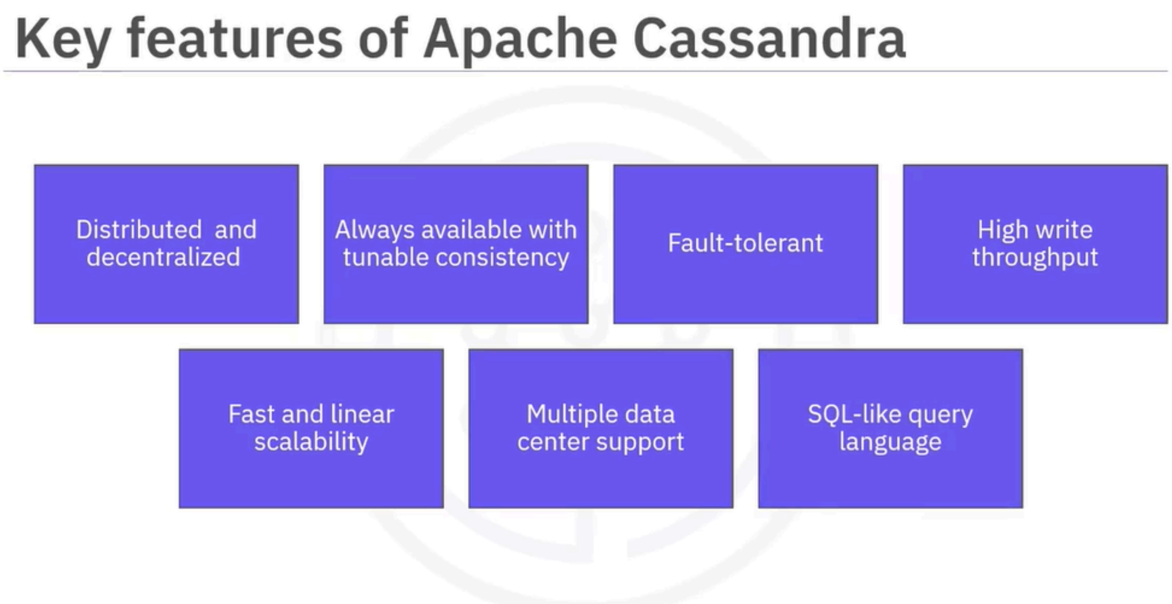
Created at Facebook, it is now used at some of the most popular sites on the Web."

Source: Carpenter, Jeff, and Eben Hewitt.
Cassandra: The Definitive Guide. O'Reilly, 2020.

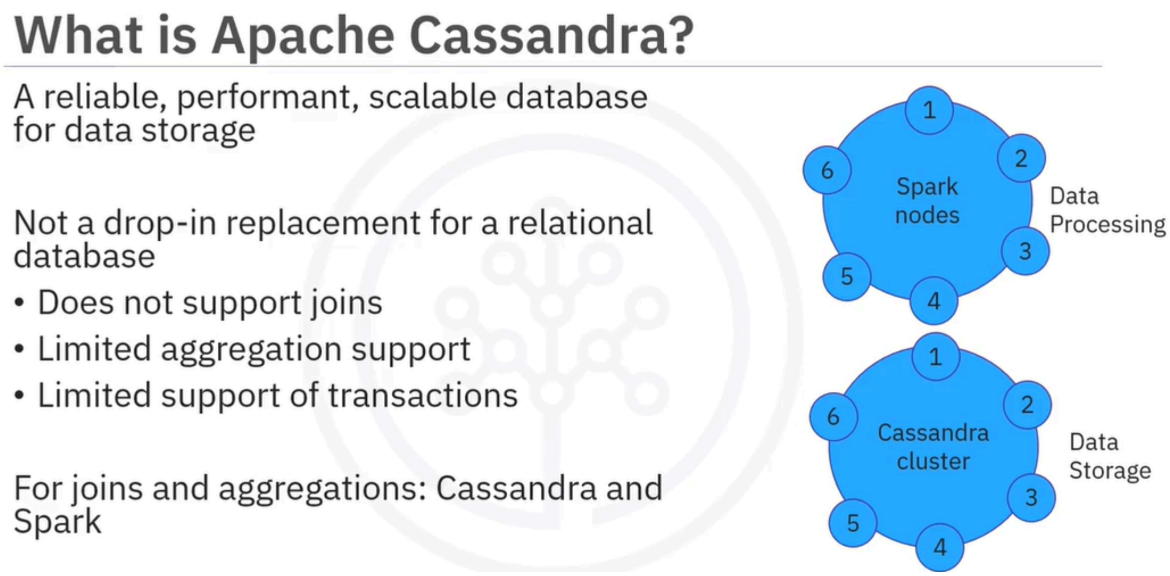
- First, let's look at what Apache Cassandra is. One of the best and most concise definitions of Cassandra has been given in the book, Cassandra: The Definitive Guide. Apache Cassandra is an open-source, distributed, decentralized, elastically scalable, highly available, fault tolerant, tunable, and consistent database that bases its distribution design on Amazon's Dynamo and its data model on Google's Bigtable. Created at Facebook, it is now used at some of the most popular sites on the web. Some of the services that use Cassandra are Netflix, Spotify, and Uber.



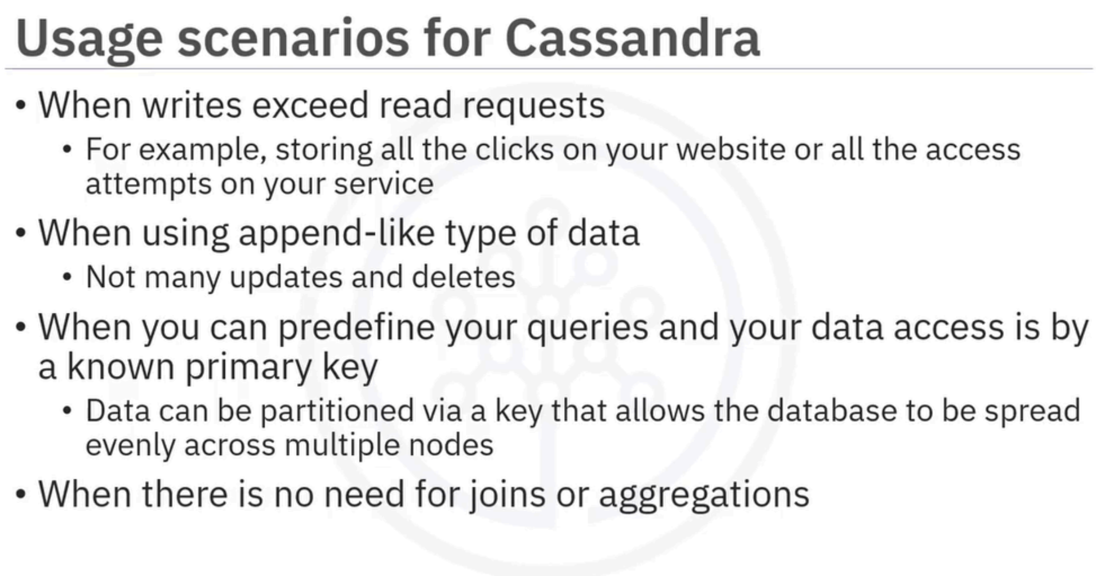
- Let's put Cassandra in a bit of perspective versus what you've learned so far about NoSQL databases, and particularly about MongoDB, the document-stored database. As you have probably noticed, NoSQL databases tend to cater to very specific use cases.
- For example, MongoDB usually covers search-related use cases where the input data can be represented as key document type of entries, but what about the use cases where you need to record some data extremely rapidly and make it available immediately for read operations, and all the while, hundreds of thousands of requests are generated? Take, for example, recording the transactions from an online shop or storing the user access info or profile for a service like Netflix.
- In this case, a solution like Apache Cassandra could be of use, while MongoDB caters to read-specific use cases, and thus is very much focused on consistency of the data, Cassandra caters to use cases that require fast storage of data, easy retrieval of data by key, availability at all times, fast scalability, and geographical distribution of the servers. One other important difference lies in the architectural choice. While MongoDB has a primary secondary architecture, Cassandra has a simpler peer to peer one.



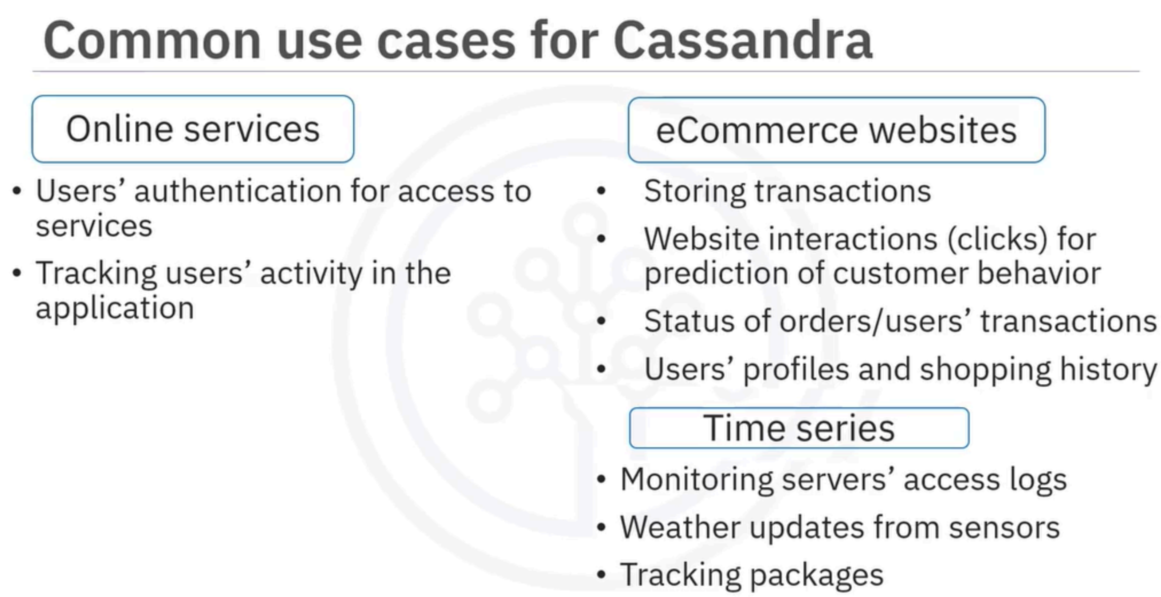
- Cassandra has a set of features that sets it apart from other NoSQL solutions: distributed and decentralized, simple peer to peer architecture, making Cassandra one of the friendliest NoSQL database installations. Always available with tunable consistency, favors availability over consistency. Fault tolerant.
- Extremely fast write throughput, while it maintains cluster performance for other operations like read. Capability to scale clusters extremely fast in a linear fashion without the need to restart or reconfigure the service, Multiple data centers deployment support, making it extremely useful for services that need to be accessed worldwide, and friendly SQL-like query language.



- Apache Cassandra is currently one of the top ten most popular solutions in the world. It's a reliable, super performant, and scalable database. Due to its popularity, Cassandra is sometimes mistaken as being a drop-in replacement for relational databases, but Cassandra by design, does not incorporate three major features of relational databases, and thus should not be seen as a drop-in replacement for a relational database. It does not support joins, has limited aggregation support, and has limited support for transactions. While writes to Cassandra are atomic, isolated, and durable in nature, the consistency part does not apply to Cassandra as there is no concept of referential integrity or foreign keys. For short, if you were thinking of using Cassandra to keep track of account balances at a bank, you probably should look at alternatives. If your application has joins and aggregations requirements, then it is best when Cassandra is paired with processing engines like Apache Spark.



- In which usage scenarios would Apache Cassandra be a good fit? When your application is write-intensive and the number of rights exceeds the number of reads. For example, storing all the clicks on your website or all the access attempts on your service. When your data doesn't have that many updates or deletes, so it comes in an append-like manner. When data access is done via a known primary key called a partition key, the key allows an even spread of the data inside the cluster and when there's no need for joins or complex aggregations as part of your queries.



- As mentioned previously, Cassandra is a best fit for globally always available types of online services and applications such as Netflix, Spotify, and Uber, but of course, there are many other use cases that can take advantage of its capabilities. Storing transactions for analytical purposes, for e-commerce websites, or the user's interactions with a website in order to personalize their experience. Just storing users profile info for services like session enrichment or granting personalized access to the service. Cassandra also thrives in time series use cases where data comes append-wise in a timely manner, like weather updates from sensors where your query could be directed towards what happened to a certain sensor in a specific time interval.

Recap

In this video, you have learned that:

- Apache Cassandra is an open-source, distributed, decentralized, elastically scalable, highly available, fault-tolerant, tunable, and consistent database
- Apache Cassandra is best used by "always available" type applications that require global distribution support, high availability, fast scalability, and high write throughputs
- Apache Cassandra is best used by online services like eCommerce websites and time series applications